

Analysis of Cryptocurrency Transaction Characteristics & Trend Prediction Based on Machine Learning

——A Case Study of BTC and ETH
Data from March to May 2025

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Content

01

Data

02

Analysis & Application

03

Conclusions & Suggestions

04

Reference



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01 Data

1.1 Data Source and Overview

1.2 Data Cleaning and Preprocessing



1.1 Data Source and Overview

- **Data Source**

1. Kaggle (<https://www.kaggle.com/>) with 50,000 2025 March-May BTC
2. ETH transaction records

- **Data Fields**

- I. Transaction ID
- II. Sender Address
- III. Receiver Address
- IV. others like Gas Price (ETH exclusive)





1.2 Data Cleaning and Preprocessing

I. Missing Value Handling

II. Timestamp Parsing

Timestamps were converted into dimensions such as date, hour, and day of the week to facilitate time series analysis

III. Duplicate Detection

No duplicate records were found in the dataset, ensuring data uniqueness

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Transaction_ID         50000 non-null  object
1   Sender_Address         50000 non-null  object
2   Receiver_Address       50000 non-null  object
3   Amount                 50000 non-null  float64
4   Transaction_Fee        50000 non-null  float64
5   Timestamp              50000 non-null  object
6   Block_ID               50000 non-null  object
7   Mining_Pool            50000 non-null  object
8   Currency               50000 non-null  object
9   Transaction_Type       50000 non-null  object
10  Transaction_Status     50000 non-null  object
11  Gas_Price_Gwei         25012 non-null  float64
dtypes: float64(3), object(9)
memory usage: 4.6+ MB
```

IV. Data Type Conversion

Categorical variables were label-encoded
numerical variables were standardized to adapt to model inputs

V. After preprocessing

50,000 valid records,
consisting of 25,012 BTC
transactions and 24,988
ETH transactions



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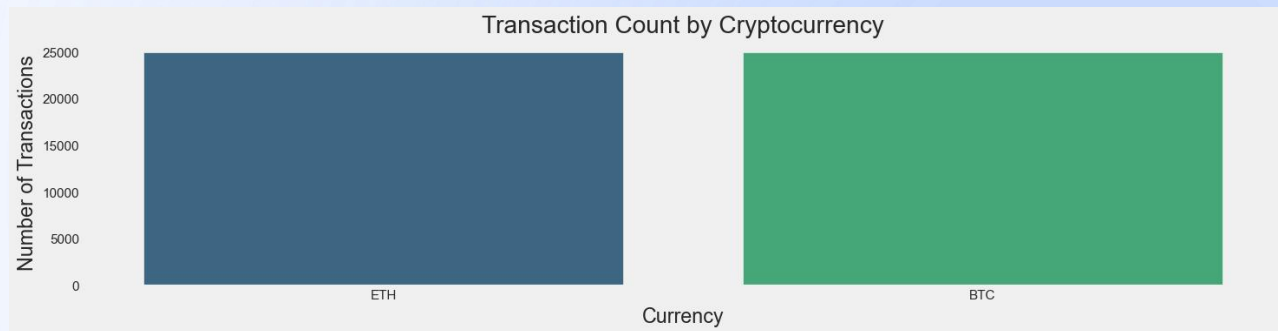
02 Analysis & Application

2.1 Exploratory Data Analysis

2.2 Model Training and Evaluation

2.1 Exploratory Data Analysis

2.1.1 Data Overview



After preprocessing

- 25,012 BTC transactions
- 24,988 ETH transactions

Transaction amount

mean = 7.5、median = 7.51、standard deviation = 4.32

Transaction Fee

the mean and median are close, standard deviation is very small

Gas_Price_Gwei

bimodal characteristic, a large standard deviation (16.40)

BTC transactions—the 25th percentile = 0

ETH transactions—the 50th percentile = 10

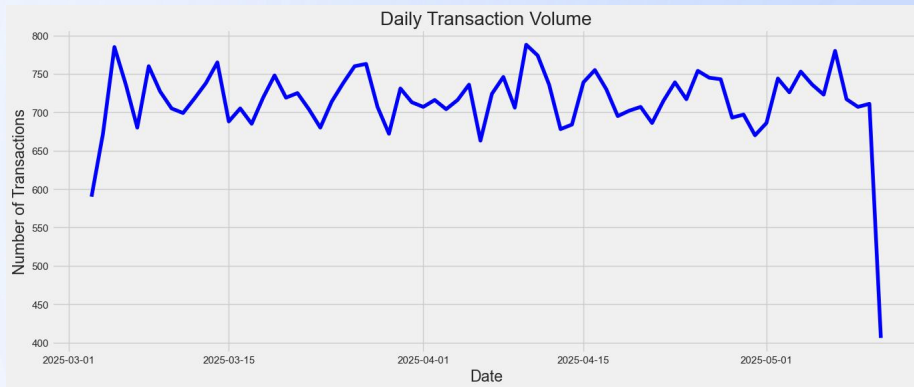
	Amount	Transaction_Fee	Gas_Price_Gwei	Hour	Day	Month	Year
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.0
mean	7.519087	0.000382	13.580300	11.419520	14.64870	3.738860	2025.0
std	4.321356	0.000257	16.403516	6.921722	8.77371	0.707471	0.0
min	0.006032	0.000010	0.000000	0.000000	1.00000	3.000000	2025.0
25%	3.784158	0.000177	0.000000	5.000000	7.00000	3.000000	2025.0
50%	7.514979	0.000340	10.000000	11.000000	14.00000	4.000000	2025.0
75%	11.234608	0.000506	25.000000	17.000000	22.00000	4.000000	2025.0
max	14.999943	0.001000	50.000000	23.000000	31.00000	5.000000	2025.0

Conclusion: the dataset is balanced regarding currency types, and no severe class imbalance issues exist.

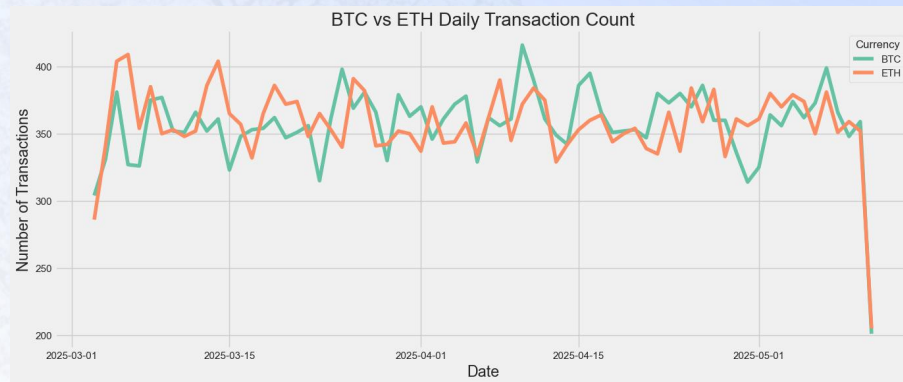


2.1 Exploratory Data Analysis

2.1.2 Transaction Trends Over Time



Daily Total Transaction Volume



Daily Transaction Trends for Bitcoin and Ethereum

- **Overall trend**

daily transaction volumes : between 600 and 800
stable

without sustained growth or decline

"jagged" fluctuations

- **Trends by currency type**

ETH exceeded BTC in mid-March and mid-April

BTC was slightly higher in late March and early May

"alternating lead"

- **Different driving factors**

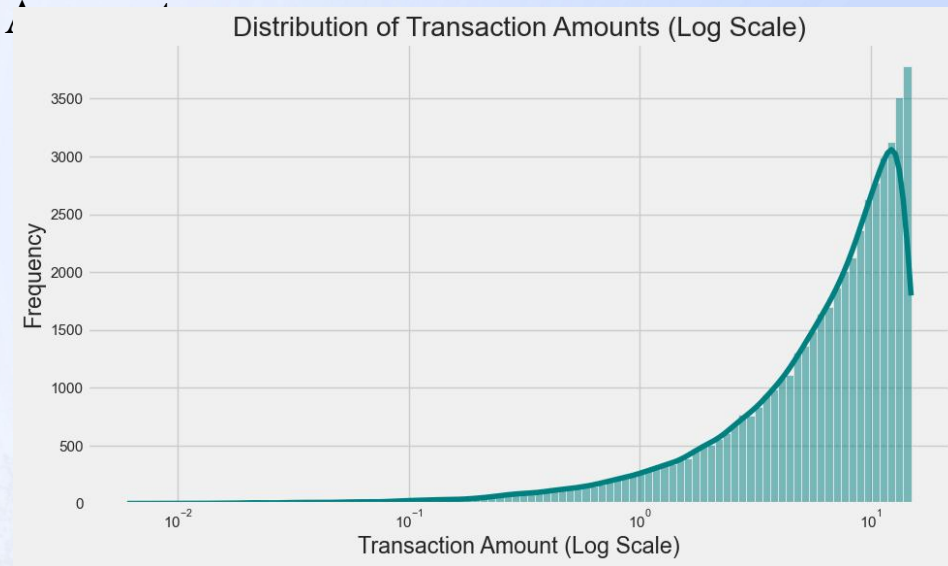
ETH : applications like DeFi and NFTs

BTC : value storage 、 cross-border payment needs



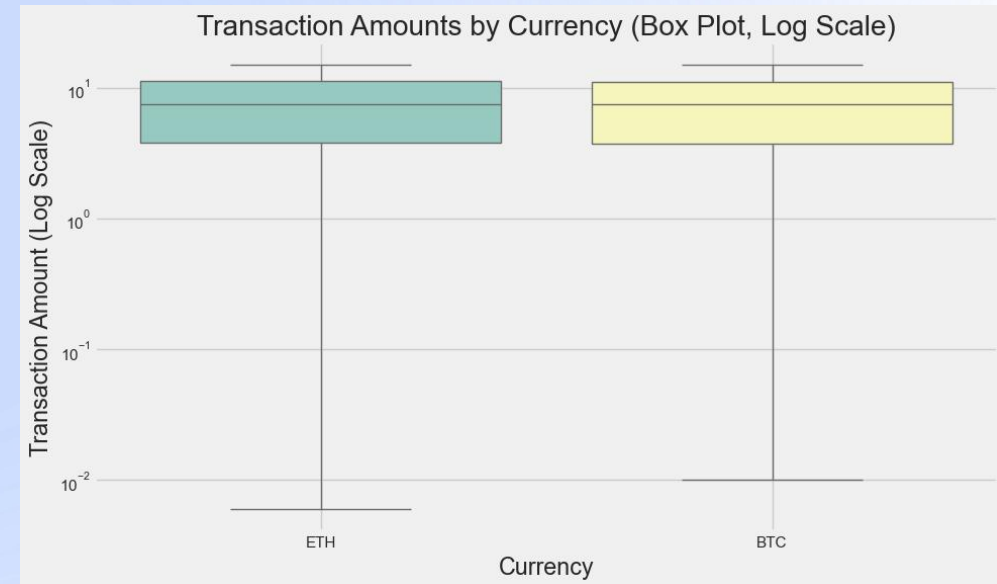
2.1 Exploratory Data Analysis

2.1.3 Analysis of Transaction Amount and Transaction Fees——Part 1: Transaction



Total Transaction Amount (Log Scale)

a right-skewed "*long tail*" distribution
most are small transactions
a few extremely large transactions ("whales")



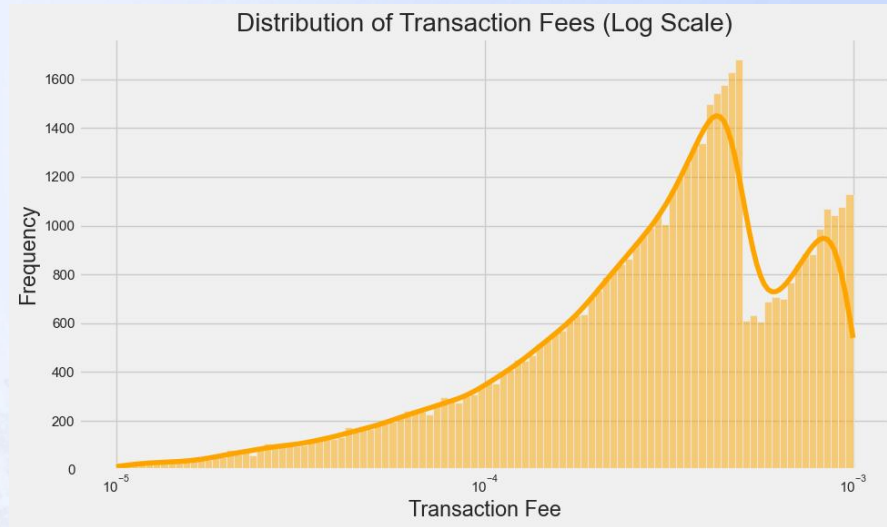
Boxplot Comparison of Transaction Amounts for Bitcoin and Ethereum

dispersion of ETH transaction amounts
slightly higher than that of BTC



2.1 Exploratory Data Analysis

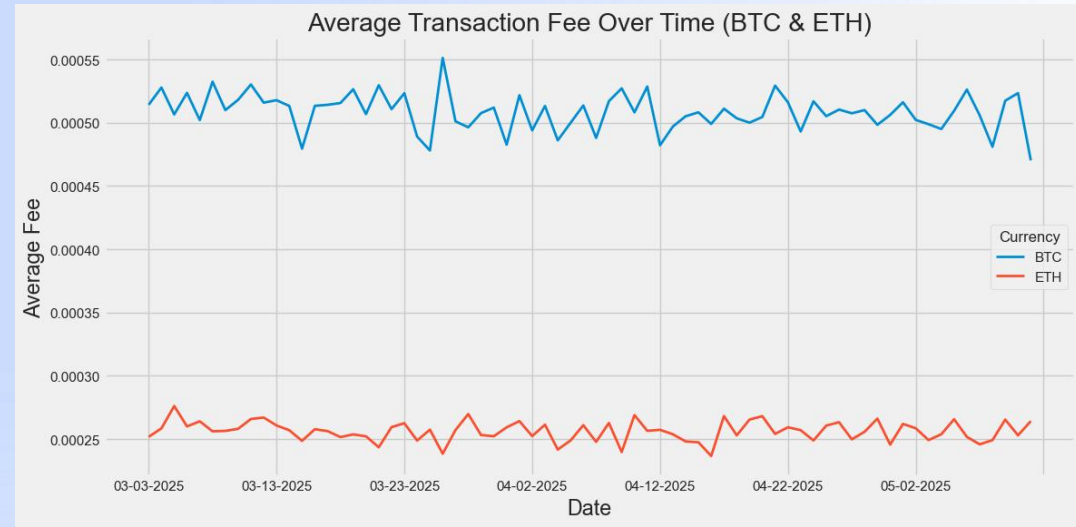
2.1.3 Analysis of Transaction Amount and Transaction Fees——Part 2: Transaction Fees



Transaction Fee Distribution (Log Scale)

A clear bimodal structure

- The distribution is skewed to the right
- The long tail part corresponds to the high Gas fees when the ETH network is congested



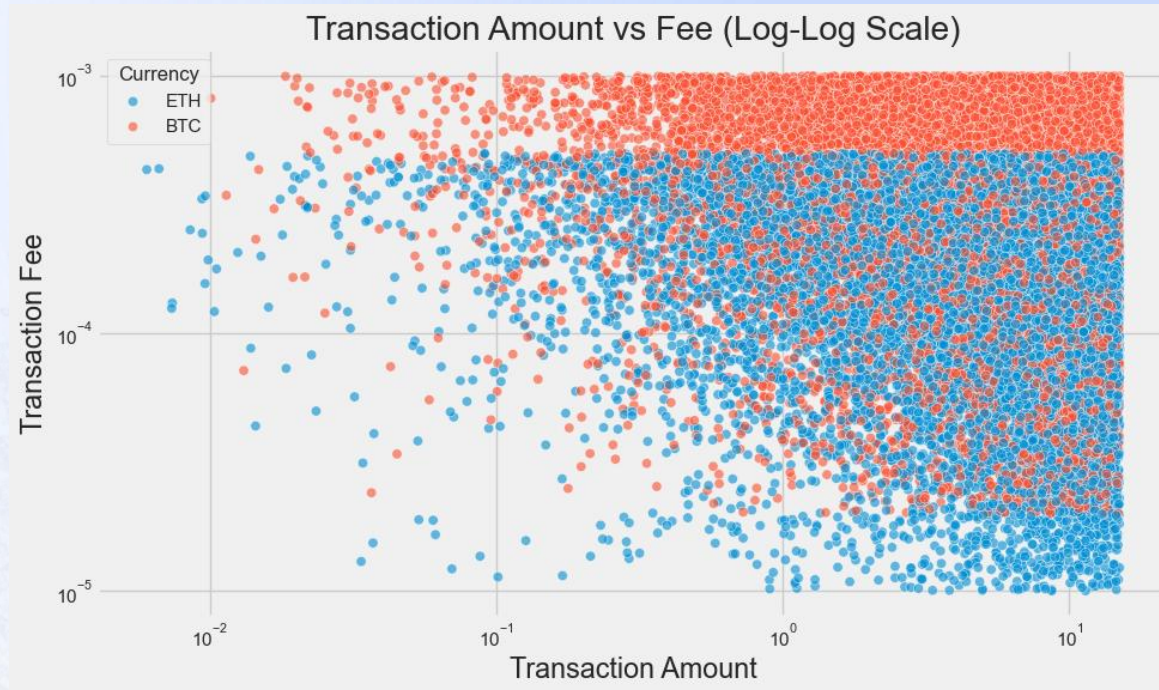
Average Transaction Fees for BTC and ETH

- BTC transactions——the first peak (extremely low transaction fees)
- ETH transactions——the second peak



2.1 Exploratory Data Analysis

2.1.3 Analysis of Transaction Amount and Transaction Fees——Part 3: Relationship



Scatter Plot of Transaction Amount vs Transaction Fee

- **No significant correlation**

- **Conclusion**

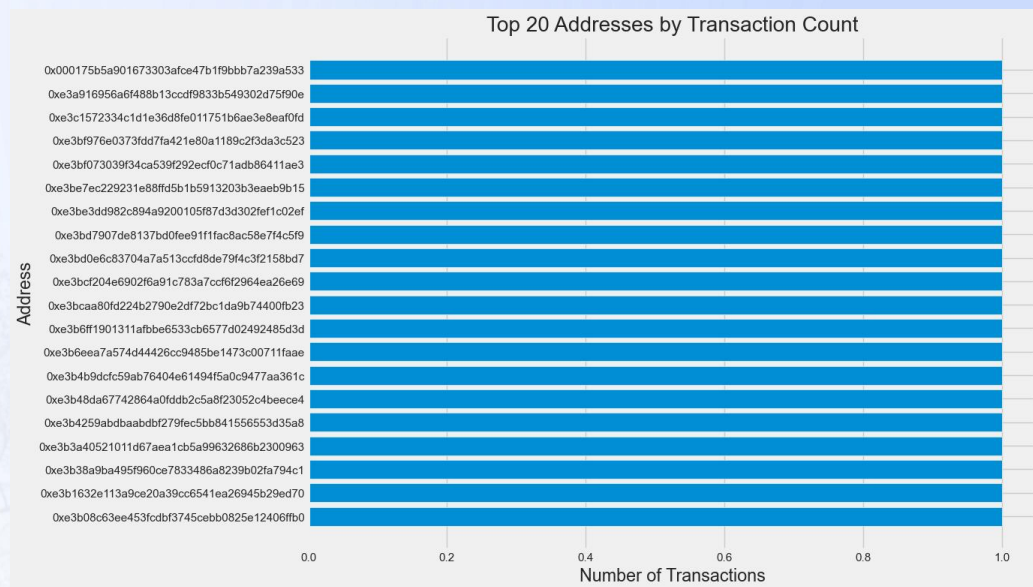
The level of handling fees is **not related to** the size of the transaction amount.

It **depends on** the network status (Gas price) and the complexity of the transaction data, rather than the value of the transfer itself.

2.1 Exploratory Data Analysis

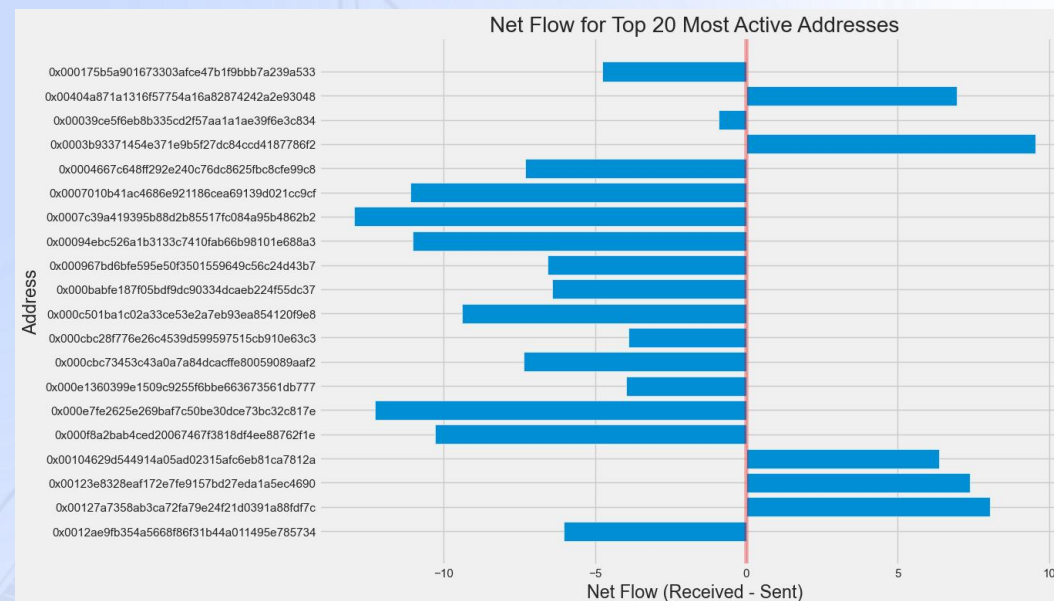
2.1.4 Analysis of Transaction Behaviors——Part 1: Frequency Analysis - Identify Frequent Traders

I. Transaction volume



- Top 20 addresses by **transaction volume**(all having only 1 transaction and identical volumes ,with no "high-frequency traders")
- highly dispersed user behavior
- No institutional investors, high-frequency arbitrageurs, or market makers present

II. Transaction amount



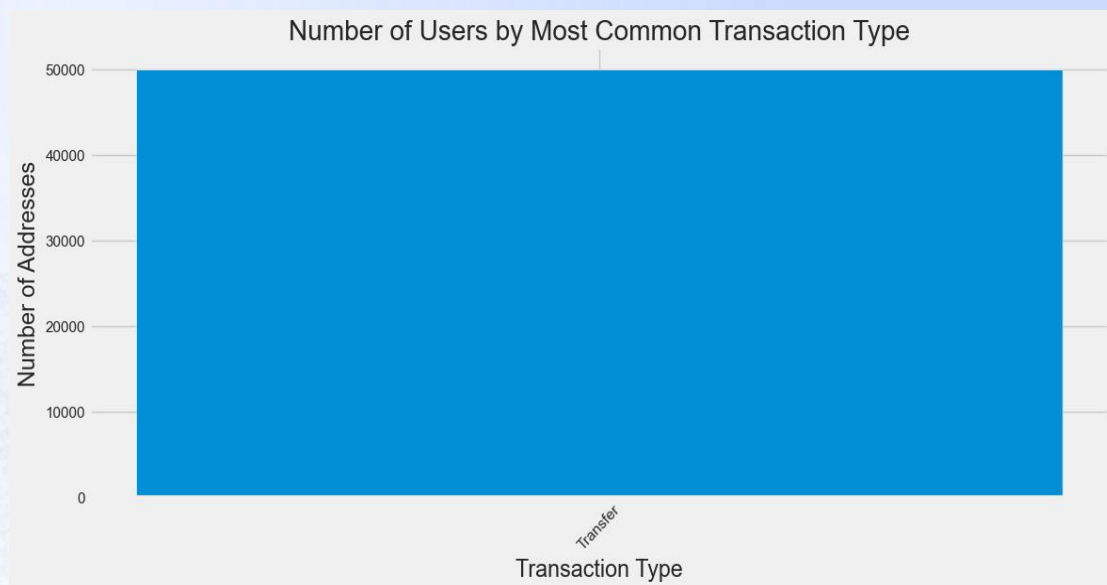
- Top 20 addresses by **net flow**
- positive and negative values distributed evenly
- singular user purposes (a one-time buy or sell)



2.1 Exploratory Data Analysis

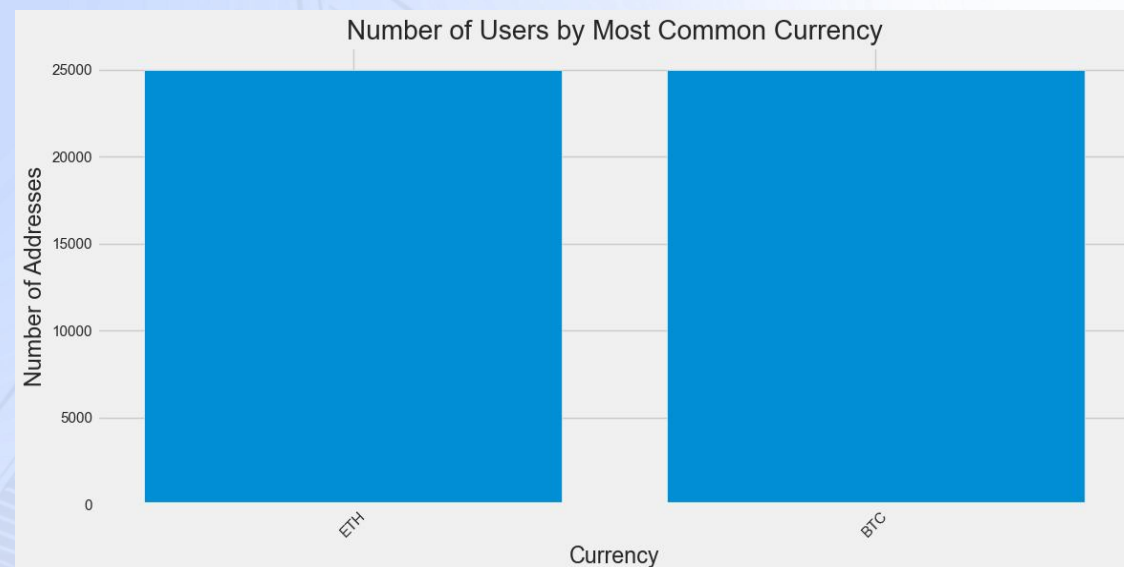
2.1.4 Analysis of Transaction Behaviors——Part 2: Analysis of User Behavior Patterns

I. The most common transaction type



"Transfer" for all addresses
with no other types

II. The most used currency



A balanced distribution——

ETH (25,012) & BTC (24,988)

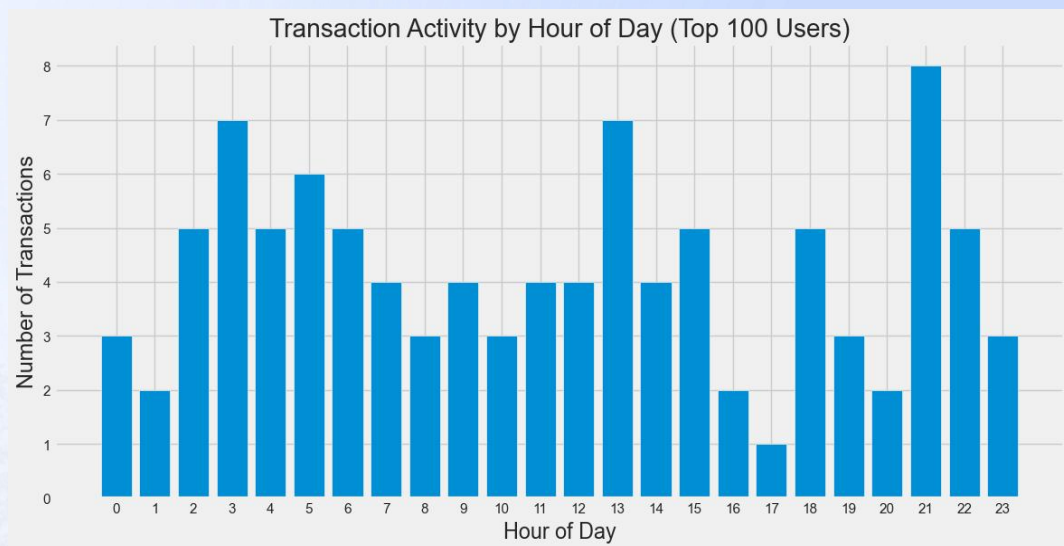
No user preference for currency



2.1 Exploratory Data Analysis

2.1.4 Analysis of Transaction Behaviors——Part 3: Time-based User Activity Patterns analysis

I. Daily

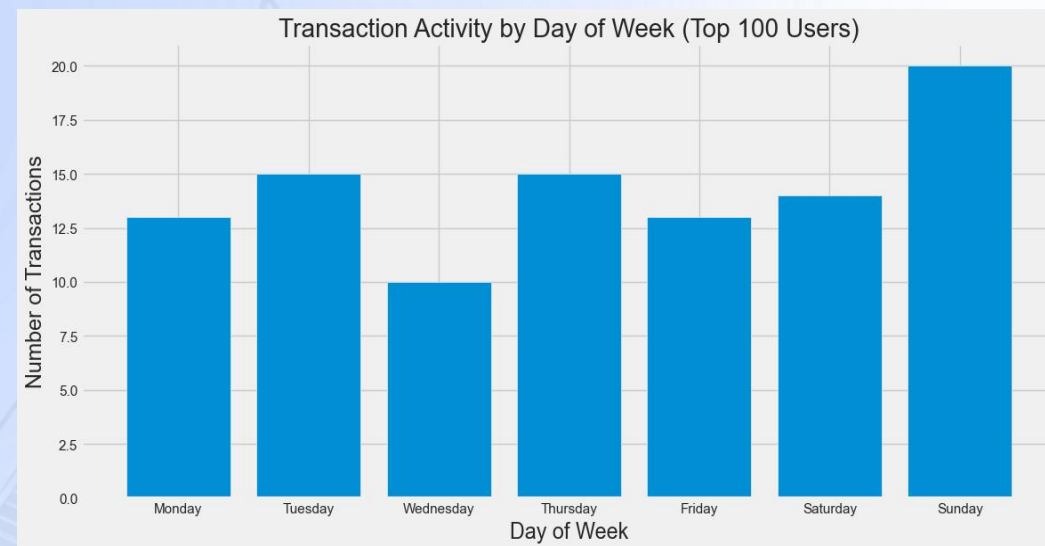


- a. With no obvious "peak hours"
- b. Hourly volumes fluctuate slightly (600-800)
- c. No concentration in trading time

Reason——

the 24/7 nature of crypto markets, allowing trading anytime

II. Weekly



- a. No obvious peaks or troughs
- b. Daily volume differences are small
- c. No "weekend effect"

Reason——

The decentralized nature of cryptocurrency frees it from traditional finance's "weekend closure" limits



2.2 Model Training and Evaluation

2.2.1 Classification

A——Objective

Predict whether the transaction is **BTC** or **ETH** based on the transaction characteristics

B——Model

Random Forest Classifier

C——Result

The accuracy rate of the test set reached 100%

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder
import pandas as pd

# Encode categorical features
categorical_cols = ['Mining_Pool', 'Transaction_Type', 'Transaction_Status']
for col in categorical_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
    df[col] = LabelEncoder().fit_transform(df[col])

# Encode target
le_currency = LabelEncoder()
df['Currency'] = le_currency.fit_transform(df['Currency']) # BTC=0, ETH=1
print(f"Currency Mapping: {dict(zip(le_currency.classes_, le_currency.transform(le_currency.classes_)))}")

# Feature selection
features = ['Amount', 'Transaction_Fee', 'Gas_Price_Gwei', 'Mining_Pool', 'Transaction_Type', 'Transaction_Status']
X = df[features]
y = df['Currency']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
clf = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    random_state=42
)
clf.fit(X_train, y_train)

# Evaluation
y_pred_train = clf.predict(X_train)
y_pred = clf.predict(X_test)
print(f"Training set accuracy rate: {accuracy_score(y_train, y_pred_train):.4f}")
print(f"Testing set accuracy rate: {accuracy_score(y_test, y_pred):.4f}")

feat_importance = pd.DataFrame({
    'feature': features,
    'importance': clf.feature_importances_
}).sort_values('importance', ascending=False)
print("\nFeature importance of classification model: ")
print(feat_importance)
```

Currency Mapping: {'BTC': np.int64(0), 'ETH': np.int64(1)}
 Training set accuracy rate: 1.0000
 Testing set accuracy rate: 1.0000

Feature importance of classification model:

	feature	importance
2	Gas_Price_Gwei	0.677383
3	Mining_Pool	0.172492
1	Transaction_Fee	0.149936
0	Amount	0.000189
4	Transaction_Type	0.000000
5	Transaction_Status	0.000000

D——Feature importance

- Gas_Price_Gwei as the Core distinguishing feature
- ETH requires Gas fees
- BTC does not



2.2 Model Training and Evaluation

2.2.2 Regression - Predicting Factors that Affect Transaction Fees

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import LabelEncoder

# Encode categorical variables
df['Currency'] = df['Currency'].fillna(df['Currency'].mode()[0])
df['Currency'] = LabelEncoder().fit_transform(df['Currency']) # BTC=0, ETH=1

cat_cols = ['Mining_Pool', 'Transaction_Type', 'Transaction_Status']
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
    df[col] = LabelEncoder().fit_transform(df[col])

# Use same features as classification plus Currency
features = ['Amount', 'Gas_Price_Gwei', 'Mining_Pool', 'Transaction_Type', 'Transaction_Status', 'Currency']
X = df[features]
y = df['Transaction_Fee']

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
reg = RandomForestRegressor(
    n_estimators=200,
    max_depth=20,
    min_samples_leaf=2,
    random_state=42
)
reg.fit(X_train, y_train)

# Evaluation
y_pred = reg.predict(X_test)
print("MSE:", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2 Score:", r2_score(y_test, y_pred))
cv_r2 = cross_val_score(reg, X, y, cv=5, scoring='r2')
print(f"Mean R2 value of 5-fold cross-validation: {cv_r2.mean():.4f} (Standard deviation: {cv_r2.std():.4f})")

feat_importance = pd.DataFrame({
    'feature': features,
    'importance': reg.feature_importances_
}).sort_values('importance', ascending=False)
print("\nImportance of model features: ")
print(feat_importance)
```

- **Objective**

- ① Predict the influencing factors of transaction fees
- ② differentiate specific weights of these factors

- **Model**

split data 80:20 for training and testing

Random Forest Regression model

- **Result**

RMSE=0.000226

low prediction error & good overall model fit

without significant deviations



2.2 Model Training and Evaluation

2.2.2 Regression - Predicting Factors that Affect Transaction Fees

- **Feature importance**

A. The most critical feature——Amount(0.35058)

transaction amount directly determines the fee level
(higher amounts usually correlate with higher fees)

MSE: 5.097955134631632e-08
RMSE: 0.00022578651719337963
R2 Score: 0.22940746129036327
Mean R² value of 5-fold cross-validation: 0.2167 (Standard deviation: 0.0083)

Importance of model features:

	feature	importance
0	Amount	0.350580
1	Gas_Price_Gwei	0.309938
5	Currency	0.307280
2	Mining_Pool	0.032202
3	Transaction_Type	0.000000
4	Transaction_Status	0.000000

B. The second most important feature——Gas_Price_Gwei(0.309938)

a core cost in blockchain transactions

C. Other features with high contribution—— Currency (0.30728)

reflecting different pricing rules and network congestion levels for different currencies

D. Other features with low or no contribution——

Mining_Pool(0.032) Transaction_Type(0) and Transaction_Status(0)



2.2 Model Training and Evaluation

2.2.3 Time Series - Predicting Daily Total Transaction Amount

```
# Forecasting Daily Total Amount

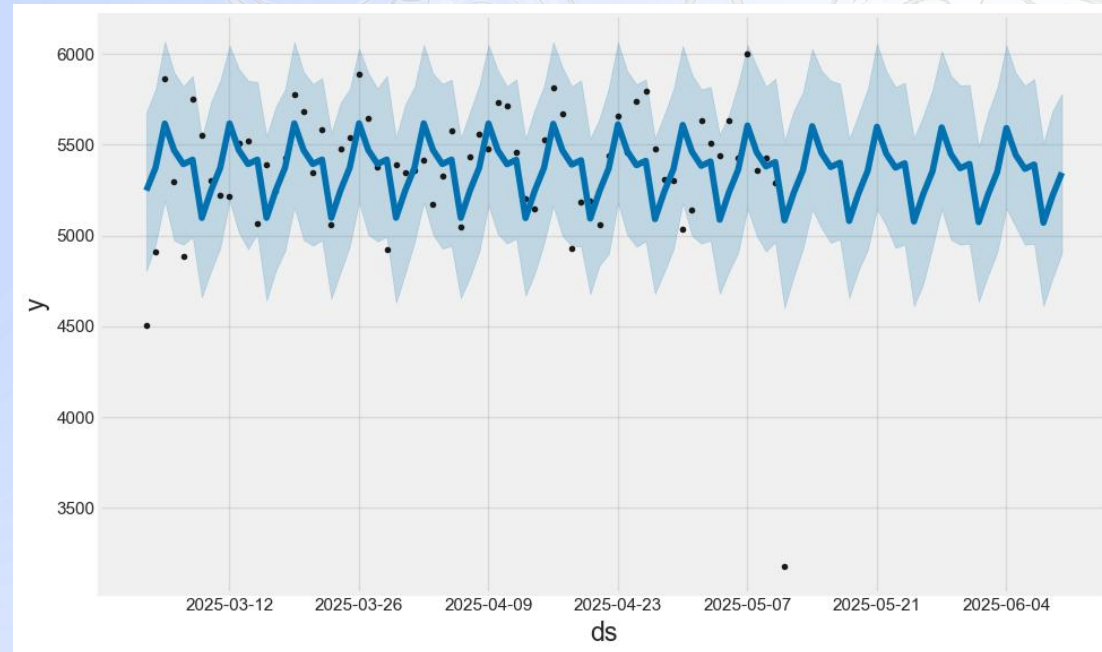
from prophet import Prophet

# Prepare data
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df_daily = df.groupby(df['Timestamp'].dt.date)['Amount'].sum().reset_index()
df_daily.columns = ['ds', 'y'] # Prophet needs these column names

# Model
model = Prophet()
model.fit(df_daily)

# Forecast
future = model.make_future_dataframe(periods=30)
forecast = model.predict(future)

# Plot
model.plot(forecast)
```



- **Objective**

Predict the total daily transaction amount for the next 30 days

- **Model**

the Prophet model

- **Result**

- ① Trend line with no dramatic slopes
- ② Trading system in a mature, stable operational state without significant external shocks
- ③ confidence interval (shaded area) is narrow
- ④ high model confidence in short-term predictions



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03 Conclusions & Suggestions

3.1 Main Conclusions

3.2 Suggestions



3.1 Main Conclusions

A——The market is mature and stable

Trading activities are continuous and uniform, possessing the robustness of a mature financial system

C——Decoupling of fees from value

Transaction fees depend on network load rather than transaction amount, reflecting the fairness of blockchain in paying based on resources and also exposing the problem of insufficient network elasticity

B——Currency mechanism and application differentiation

BTC and ETH have significant differences in fee mechanisms and application scenarios, with Gas prices being the core differentiating feature

D——Decentralized user behavior

The market is dominated by low-frequency and discrete retail users, and there are no obvious signs of institutional manipulation



3.2 Suggestions

Investors

Optimize Trading Timing and Cost Management

- ✓ **Avoid Network Congestion to Reduce Costs**

Take advantage of the 7x24-hour feature to conduct large non-urgent transactions during periods of low network load to save expenses

- ✓ **Rational Asset Allocation**

Choose BTC for value storage when participating in on-chain ecosystems, bear the risk of Gas fluctuations of ETH and consider the Layer 2 solution

Blockchain Developers & Network Operators

Enhance Network Elasticity and Scaling Solutions

- ✓ **Targeted Scaling Implementation**

Accelerate the implementation of technologies such as Layer 2 and sharding to smooth out the peak Gas price during congestion

- ✓ **Optimize Micropayment Experience**

Optimize the experience for small transactions of BTC (such as the Lightning Network)

Regulatory Bodies

Establish Dynamic and Penetrating Regulatory Frameworks

- ✓ **Implement penetrating supervision**

Establish cross-address fund flow monitoring and focus on extremely large transactions in the "long tail"

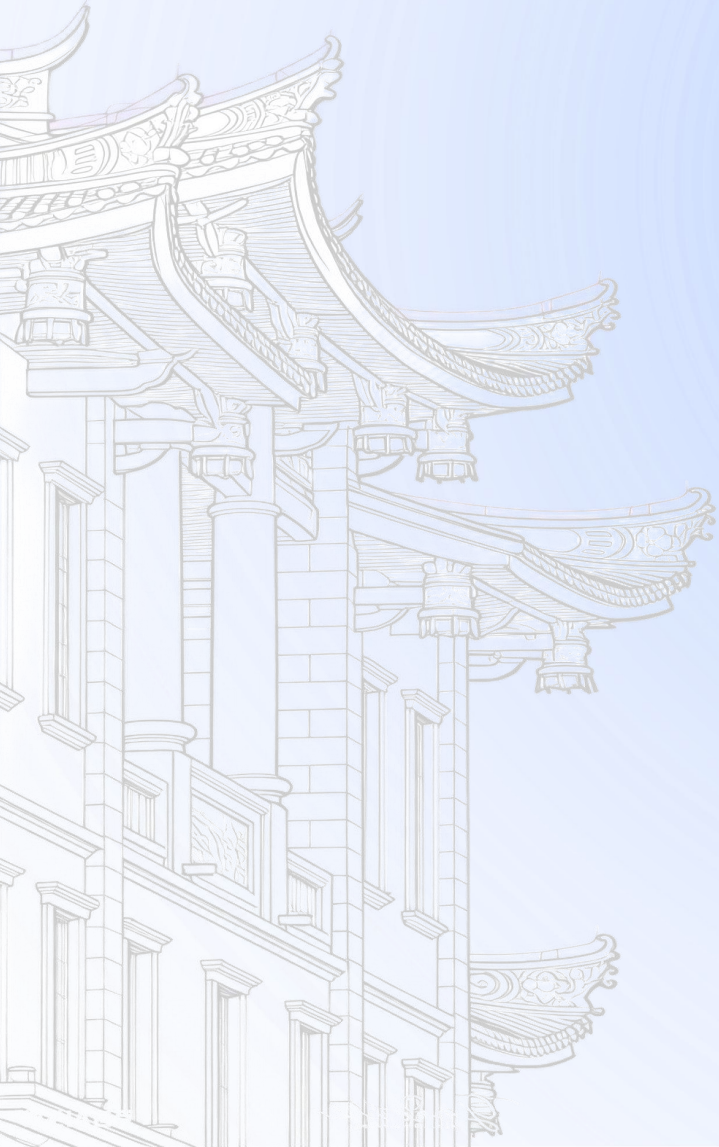
- ✓ **Adapt Regulatory Mechanisms to 24/7 Markets**

Establish a 7x24-hour real-time dynamic monitoring system by using AI tools



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04 Reference

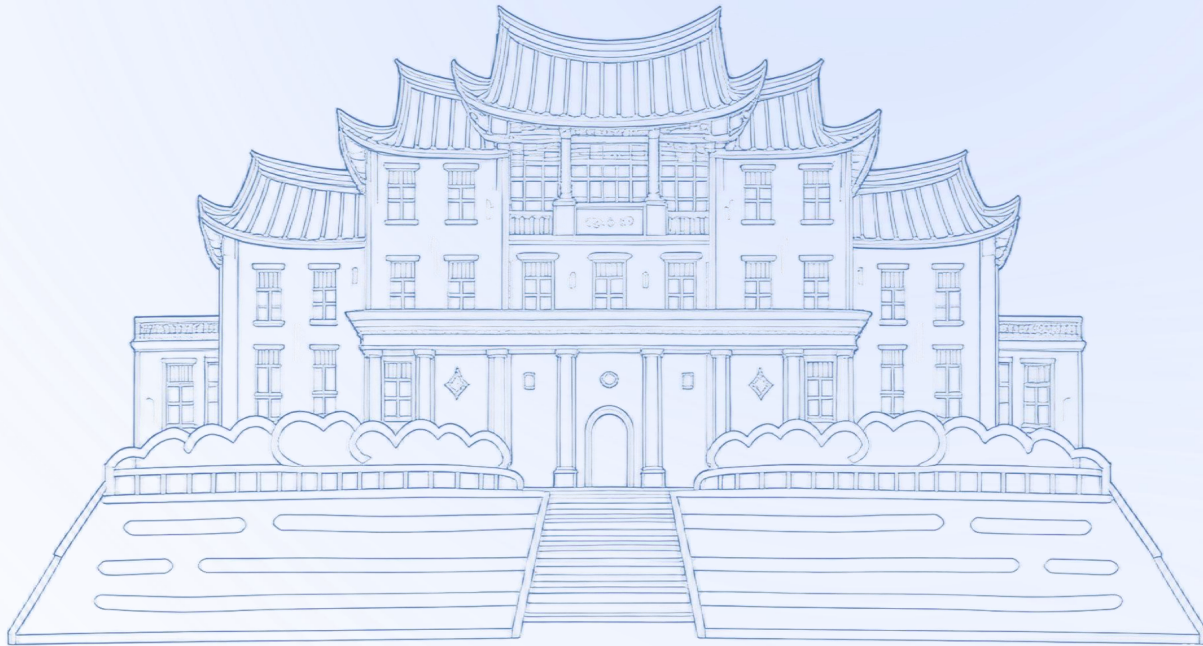




Reference

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- [6]Nazir S ,Mehmmmod S ,Junxia L , et al.RETRACTED: Supply chain efficiency via knowledge management and blockchain adoption readiness[J].Journal of Manufacturing Technology Management,2026,37(3):704-728.
- [7]Singh, A., et al. (2025). A review of mathematical models for pricing, risk, and optimization in cryptocurrency analytics. Journal of Digital Economy, 4, 100-125.

Thanks for watching!



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